Restaurant Performance

Problem Statement

The Business Context

Zomato is one of India's largest food discovery and restaurant aggregation platforms. It operates across thousands of restaurants in Bangalore, and its core value to customers is helping them **discover restaurants, order food online, reserve tables, and read/share reviews**.

The core problem the management team has noticed is this: **despite having a huge number of partner restaurants on the platform, performance is very uneven.** Some restaurants receive excellent ratings and thousands of customer votes — meaning they're clearly thriving, well-loved, and highly engaged with. Others, even when they're located in prime, high-traffic areas, struggle to attract any meaningful engagement at all — few votes, inconsistent ratings, low visibility.

This unevenness is the actual mystery the business wants solved. Zomato's management doesn't just want to know "which restaurants are good" — they want to understand **why** some succeed and others don't, so they can act on it.

Your Role

You are cast as a **Data Analyst** hired to dig into the restaurant dataset and produce **actionable insights** — meaning not just charts and numbers, but conclusions the business can actually *act on* to:

- Improve customer satisfaction
- Strengthen restaurant partnerships
- Support strategic business decisions

The Project Objective (the one-sentence mission)

"Perform an end-to-end exploratory and business analysis of Bangalore restaurants listed on Zomato to identify the major factors influencing restaurant success and provide data-driven recommendations."

Notice the phrase "end-to-end" — this means the project isn't just EDA charts in isolation, it's meant to flow from raw data all the way to final business recommendations, which is exactly the phase-by-phase structure we've been following (Cleaning → EDA → Feature Engineering → Analysis → Dashboard → Recommendations).

The 10 Business Questions (the actual things you must answer)

These are the specific questions the business wants answered, and you've now actually answered every single one through your EDA work:

1. **Which restaurants receive the highest ratings and customer votes?** — identifying top performers
2. **Are ratings correlated with customer engagement?** — you found: moderately (0.4), with the "funnel" pattern — high engagement almost guarantees decent ratings, but low engagement is unpredictable
3. **Does online ordering improve popularity?** — you found: no meaningful effect (3.59 vs 3.65 rating)
4. **Does table booking influence ratings?** — you found: yes, a real gap (3.57 vs 4.11), though likely a proxy for premium dining rather than a direct cause
5. **Which restaurant types perform best?** — you found: Bar/Pub/Fine Dining combinations outperform the dominant Quick Bites/Casual Dining categories
6. **Which Bangalore locations have the highest restaurant concentration?** — you found: tech corridors (Whitefield, Electronic City, BTM, HSR, Marathahalli)
7. **Which locations have the highest average ratings?** — you found: central Bangalore (Lavelle Road, St. Marks Road, Church Street), with Koramangala 5th Block standing out as both high-volume and high-quality
8. **Which cuisines are most popular and best rated?** — you found: North Indian/Chinese/South Indian dominate by volume, but niche international cuisines (Japanese, Mediterranean, European) rate higher
9. **Does pricing influence customer satisfaction?** — you found: a modest positive relationship (0.33), premium pricing acts as a "quality floor"
10. **Which market segments have the highest growth potential?** — this is the synthesis question — your cross-cutting finding (premium/niche segments consistently outperform saturated mass-market segments across type, location, and cuisine) is your answer here

The 5 Expected Deliverables

1. **Data Cleaning** — duplicates, missing values, rating cleaning, cost conversion, cuisine standardization (✅ done)
2. **Exploratory Data Analysis** — summary statistics, distributions, correlations, outliers (✅ done)
3. **Business Analysis** — covering customer behavior, restaurant performance, pricing, cuisine, location, online ordering, table booking (✅ done, via EDA)

4. **Interactive Dashboard in Tableau or Power BI** — with KPI cards, bar charts, maps, treemaps, scatter plots, heatmaps, filters, and rankings (□ next step — what we're about to build)
5. **Final recommendations** — for restaurant expansion, pricing, promotions, cuisine strategy, and customer engagement (□ comes after the dashboard)

The Business Impact (why this matters, ultimately)

The stated goal is that this analysis should help Zomato:

- Improve restaurant recommendations shown to customers
- Identify high-performing locations and cuisines
- Optimize marketing campaigns
- Strengthen restaurant partnerships
- Support expansion strategies
- Enable data-driven decision making overall